

Data availability

PolyCrit can be found at <https://lccc.ywangcomp.org>.

References

- [1] T. Chang, Recent Advances in Liquid Chromatography Analysis of Synthetic Polymers, Springer Berlin Heidelberg, Berlin, Heidelberg, 2003, pp. 1–60.
- [2] T. Macko, R. Brüll, Y. Zhu, Y. Wang, A review on the development of liquid chromatography systems for polyolefins, *J. Sep. Sci.* 33 (22) (2010) 3446–3454.
- [3] D.J. Audus, J.J. de Pablo, Polymer informatics: opportunities and challenges, *ACS Macro Lett.* 6 (10) (2017) 1078–1082.
- [4] W. Radke, Polymer separations by liquid interaction chromatography: Principles – prospects – limitations, *J. Chromatogr. A* 1335 (2014) 62–79.
- [5] D. Berek, Coupled liquid chromatographic techniques for the separation of complex polymers, *Prog. Polym. Sci.* 25 (7) (2000) 873–908.
- [6] Y. Wang, Theory and Principles of Interaction Chromatography, ACS Symposium Series, pp. 19–30, ch. 2.
- [7] E.F. Casassa, Theoretical models for peak migration in gel permeation chromatography, *J. Phys. Chem.* 75 (26) (1971) 3929–3939.
- [8] A.A. Gorbunov, A.M. Skvortsov, Statistical properties of confined macromolecules, *Adv. Colloid Interface Sci.* 62 (1) (1995) 31–108.
- [9] Y. Gong, Y. Wang, Partitioning of polymers into pores near the critical adsorption point, *Macromol.* 35 (19) (2002) 7492–7498.
- [10] S. Orelli, W. Jiang, Y. Wang, A computational investigation of the critical condition used in the liquid chromatography of polymers, *Macromol.* 37 (26) (2004) 10073–10078.
- [11] Y. Wang, A. Masur, Y. Zhu, J. Ziebarth, Partitioning of star branched polymers into pores at three chromatography conditions, *J. Chromatogr. A* 1217 (39) (2010) 6102–6109.
- [12] T. Macko, D. Hunkeler, Liquid Chromatography under Critical and Limiting Conditions: A Survey of Experimental Systems for Synthetic Polymers, Springer Berlin Heidelberg, Berlin, Heidelberg, 2003, pp. 62–136.
- [13] A. Baumgaertel, E. Altuntaş, U.S. Schubert, Recent developments in the detailed characterization of polymers by multidimensional chromatography, *J. Chromatogr. A* 1240 (2012) 1–20.
- [14] B. Trathnigg, M.I. Malik, O. Jamelnik, N.V. Cuong, A data base for polymer chromatography: Dependence of interaction parameters on mobile phase composition, *Anal. Chim. Acta* 604 (1) (2007) 39–44.
- [15] S. Abbott, C.M. Hansen, Hansen Solubility Parameters in Practice, Hansen Solubility, 2008.
- [16] C.M. Hansen, Hansen Solubility Parameters: A User's Handbook, CRC Press, 2007.
- [17] S. Choi, J. Lee, J. Seo, S.W. Han, S.H. Lee, J.-H. Seo, J. Seok, Automated BigSMILES conversion workflow and dataset for homopolymeric macromolecules, *Sci. Data* 11 (1) (2024) 371.
- [18] P. Nefedov, T. Zhmakina, Effect of temperature, solvent composition and pressure on the adsorption and chromatography of polystyrenes on macroporous glasses, *Polym. Sci. U. S. S. R.* 23 (2) (1981) 304–312.
- [19] S.T. Balke, R.D. Patel, Orthogonal chromatography, 281–310, ch. 16.
- [20] A.V. Gorskov, V.V. Jevreinov, B. Lausecker, H. Pasch, H. Becker, G. Wagner, Flüssigchromatographie von makromolekülen unter kritischen Bedingungen. Zur auftrennung von poly(oxypropylenpolyolen) nach ihrer funktionalität, *Acta Polym.* 37 (11–12) (1986) 740–741.
- [21] A. Gorskov, T. Prudskova, V. Gur'yanova, V. Evreinov, Functional type separation of oligocarbonates by liquid chromatography under critical conditions: Effects of pore size, *Polym. Bull.* 15 (5) (1986).
- [22] V. Gur'yanova, A. Pavlov, Step-by-step analysis of polycondensation oligomers and polymers by liquid chromatography, *J. Chromatogr. A* 365 (1986) 197–204.
- [23] A. Gorskov, H. Much, H. Becker, H. Pasch, V. Evreinov, S. Entelis, Chromatographic investigations of macromolecules in the 'critical range' of liquid chromatography, *J. Chromatogr. A* 523 (1990) 91–102.
- [24] H. Krüger, H. Much, G. Schulz, G. Wehrstedt, Determination of functionality and molecular weight distributions in α -hydroxy- ω -benzyloxyoligo(1,3,6-trioxane)s using HPLC methods, *Die Makromol. Chem.* 191 (4) (1990) 907–920.
- [25] T. Zimina, Y. Kever, Y. Melenevskaya, V. Zgonnik, B. Belen'kii, Experimental verification of the concept of chromatographic 'invisibility' in critical chromatography of block copolymers, *Polym. Sci. U. S. S. R.* 33 (6) (1991) 1250–1254.
- [26] H. Pasch, C. Brinkmann, H. Much, U. Just, Chromatographic investigations of macromolecules in the 'critical range' of liquid chromatography, *J. Chromatogr. A* 623 (2) (1992) 315–322.
- [27] T. Zimina, J. Kever, E. Melenevskaya, A. Fell, Analysis of block copolymers by high-performance liquid chromatography under critical conditions, *J. Chromatogr. A* 593 (1–2) (1992) 233–241.
- [28] T.M. Zimina, A.F. Fell, J.B. Castledine, Diode-array detection with spectral suppression in size exclusion and adsorption chromatography for rapid evaluation of chemical composition in block copolymers of styrene and methyl methacrylate, *Polymer* 33 (19) (1992) 4129–4135.
- [29] H. Pasch, C. Brinkmann, Y. Gallot, Chromatographic investigations of macromolecules in the critical range of liquid chromatography: 4. Analysis of poly(styrene-*b*-methyl methacrylate), *Polymer* 34 (19) (1993) 4100–4104.
- [30] H. Pasch, M. Augenstein, Chromatographic investigations of macromolecules in the critical range of liquid chromatography: 5. Characterization of block copolymers of decyl and methyl methacrylate, *Die Makromol. Chem.* 194 (9) (1993) 2533–2541.
- [31] H. Pasch, Chromatographic investigations of macromolecules in the critical range of liquid chromatography: 3. Analysis of polymer blends, *Polymer* 34 (19) (1993) 4095–4099.
- [32] H. Pasch, Y. Gallot, B. Trathnigg, Chromatographic investigations of macromolecules in the critical range of liquid chromatography: 7. Analysis of the poly(methyl methacrylate) block in poly(styrene-block-methyl methacrylate), *Polymer* 34 (23) (1993) 4986–4989.
- [33] B. Trathnigg, D. Thamer, X. Yan, S. Kinugasa, Analysis of polyethers by isocratic HPLC with universal detectors. I. optimization of chromatographic conditions, *J. Liq. Chromatogr.* 16 (12) (1993) 2439–2452.
- [34] P.J.C.H. Cools, A.M. Van Herk, A.L. German, W. Staal, Critical retention behavior of homopolymers, *J. Liq. Chromatogr.* 17 (14–15) (1994) 3133–3143.
- [35] R.-P. Krüger, H. Much, G. Schulz, Determination of functionality and molar mass distribution of aliphatic polyesters by orthogonal liquid chromatography. I. Off-line investigation of poly(1,6-hexanediol adipates), *J. Liq. Chromatogr.* 17 (14–15) (1994) 3069–3090.
- [36] H. Pasch, M. Augenstein, B. Trathnigg, Chromatographic investigations of macromolecules in the critical range of liquid chromatography: 6. Analysis of the methyl methacrylate block in block copolymers of decyl and methyl methacrylate, *Macromol. Chem. Phys.* 195 (2) (1994) 743–750.
- [37] H. Pasch, I. Zammert, Chromatographic investigations of macromolecules in the critical range of liquid chromatography. VIII. Analysis of polyethylene oxides, *J. Liq. Chromatogr.* 17 (14–15) (1994) 3091–3108.
- [38] B. Trathnigg, D. Thamer, X. Yan, B. Maier, H.-R. Holzbauer, H. Much, Characterization of ethoxylated fatty alcohols using liquid chromatography with density and refractive index detection II. Quantification in liquid chromatography under critical conditions, *J. Chromatogr. A* 665 (1) (1994) 47–53.
- [39] P. Kilz, R.-P. Krüger, H. Much, G. Schulz, Two-dimensional chromatography for the deformation of complex copolymers, 1995, pp. 223–241.
- [40] H. Pasch, K. Rode, Use of matrix-assisted laser desorption/ionization mass spectrometry for molar mass-sensitive detection in liquid chromatography of polymers, *J. Chromatogr. A* 699 (1–2) (1995) 21–29.
- [41] V.B. Gancheva, N.G. Vladimirov, R.S. Velichkova, Chromatographic investigations of polyacetals in the critical range of liquid chromatography, 1. Analysis of 9-anthrylmethyl-terminated poly(1,3,6-trioxacyclooctane)s, *Macromol. Chem. Phys.* 197 (5) (1996) 1757–1770.
- [42] R. Krüger, H. Much, G. Schulz, O. Wachsen, New aspects of determination of polymer heterogeneity by 2-Dimensional orthogonal liquid chromatography and maldi-tof-MS, *Macromol. Symp.* 110 (1) (1996) 155–176.
- [43] R.P. Krüger, H. Much, G. Schulz, Determination of polymer heterogeneity by two-dimensional orthogonal liquid chromatography, *Int. J. Polym. Anal. Charact.* 2 (3) (1996) 221–235.
- [44] H.C. Lee, T. Chang, Characterization of binary polymer mixtures by simultaneous size exclusion chromatography and interaction chromatography, *Macromol.* 29 (22) (1996) 7294–7296.
- [45] H. Pasch, K. Rode, Chromatographic investigations of macromolecules in the critical range of liquid chromatography, 10. Polymer blend analysis using a viscosity detector, *Macromol. Chem. Phys.* 197 (9) (1996) 2691–2701.
- [46] H. Pasch, K. Rode, N. Chaumien, Chromatographic investigations of macromolecules in the critical range of liquid chromatography: 9. Separation of methacrylate-based polymer blends, *Polymer* 37 (18) (1996) 4079–4083.
- [47] H. Pasch, A. Deffieux, I. Henze, M. Schappacher, L. Rique-Lurbet, Analysis of macrocyclic polystyrenes. 1. Liquid chromatographic investigations, *Macromol.* 29 (27) (1996) 8776–8782.
- [48] H. Philipsen, B. Klumperman, A. van Herk, A. German, Critical retention behaviour of polymers a study on the influence of some practical parameters, *J. Chromatogr. A* 727 (1) (1996) 13–25.
- [49] D. Berek, M. Janco, K. Hatada, T. Kitayama, N. Fujimoto, Separation of poly(methyl methacrylate)s according to their tacticity II. Chromatographic investigations of poly(methyl methacrylate)s with different tacticity at the critical adsorption point, *Polym. J.* 29 (12) (1997) 1029–1033.
- [50] T. Chang, H.C. Lee, W. Lee, High performance liquid chromatography characterization of macromolecules, *Macromol. Symp.* 118 (1) (1997) 261–265.
- [51] G. Czichocki, R. Heger, W.A. Goedel, H. Much, Liquid chromatography of polyisoprenes: determination of critical composition using non protic binary and acidic ternary solvent mixtures, *J. Chromatogr. A* 791 (1–2) (1997) 350–356.
- [52] M. Janco, D. Berek, A. Önen, C.H. Fischer, Y. Yagci, W. Schnabel, Separation of block copolymers from parent homopolymers by means of liquid chromatography at the critical adsorption point, *Polym. Bull.* 38 (6) (1997) 681–688.